

Physics for Engineers 2 Lab

BUOYANCY

MPS Department | FEU Institute of Technology



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OBJECTIVES

- To determine the buoyant force of an object using water displacement method
- To determine the buoyant force of an object using loss of mass method



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Buoyancy

Buoyancy is the ability of the object to float in fluid (gas or liquid)

Buoyant force is the upward force acted on a partially or fully submerged object. It acts opposite the gravitational force.

In the following experiment, two ways of determining the buoyant force are presented: the fluid displacement method and the loss of mass method>



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Fluid Displacement Method

- In this method, the block is fully submerged in different types of fluid and the fluid is displaced. The volume of the fluid displaced is the same as the volume of the block.

$$V_{displaced} = V_{block}$$

- The buoyant force experienced by the block is

$$B_1 = W_{fluid\ displaced} = m_{fluid\ displaced} \cdot g = \rho_{medium} \cdot g \cdot V_{displaced}$$
$$B_1 = \rho_{medium} \cdot V_{displaced} \cdot g$$

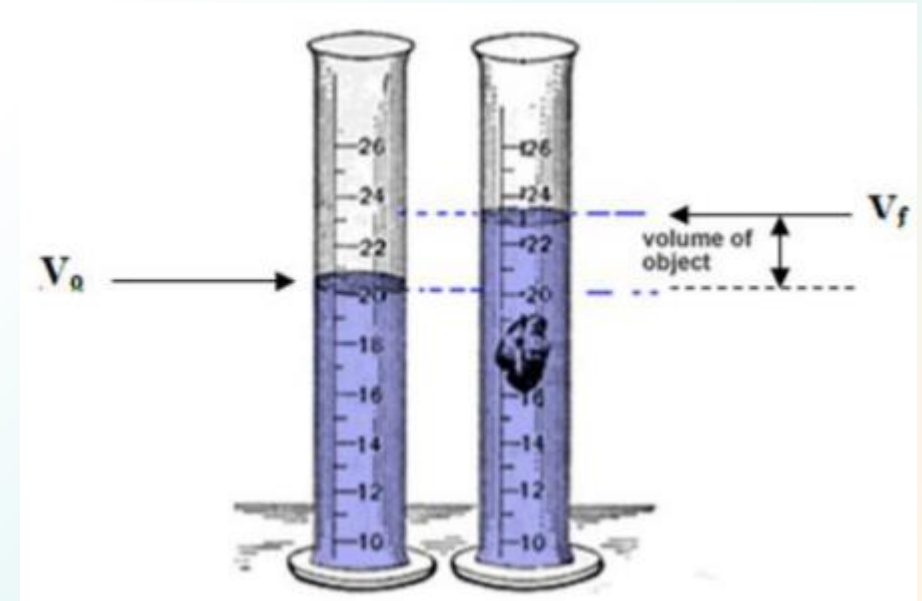
where

B_1 is the buoyant force

ρ_{medium} is the density of the fluid

$V_{displaced}$ is the volume of the fluid displaced

g is the gravitational acceleration, $g = 9.80 \text{ m/s}^2$



Loss of Mass Method

- In this method, the block is weighed in air and then while it is fully submerged in different types of fluid. The apparent weight of the block, that is, when it is submerged is lower than the its weight in air. This is due to buoyant force
- The buoyant force experienced by the block is

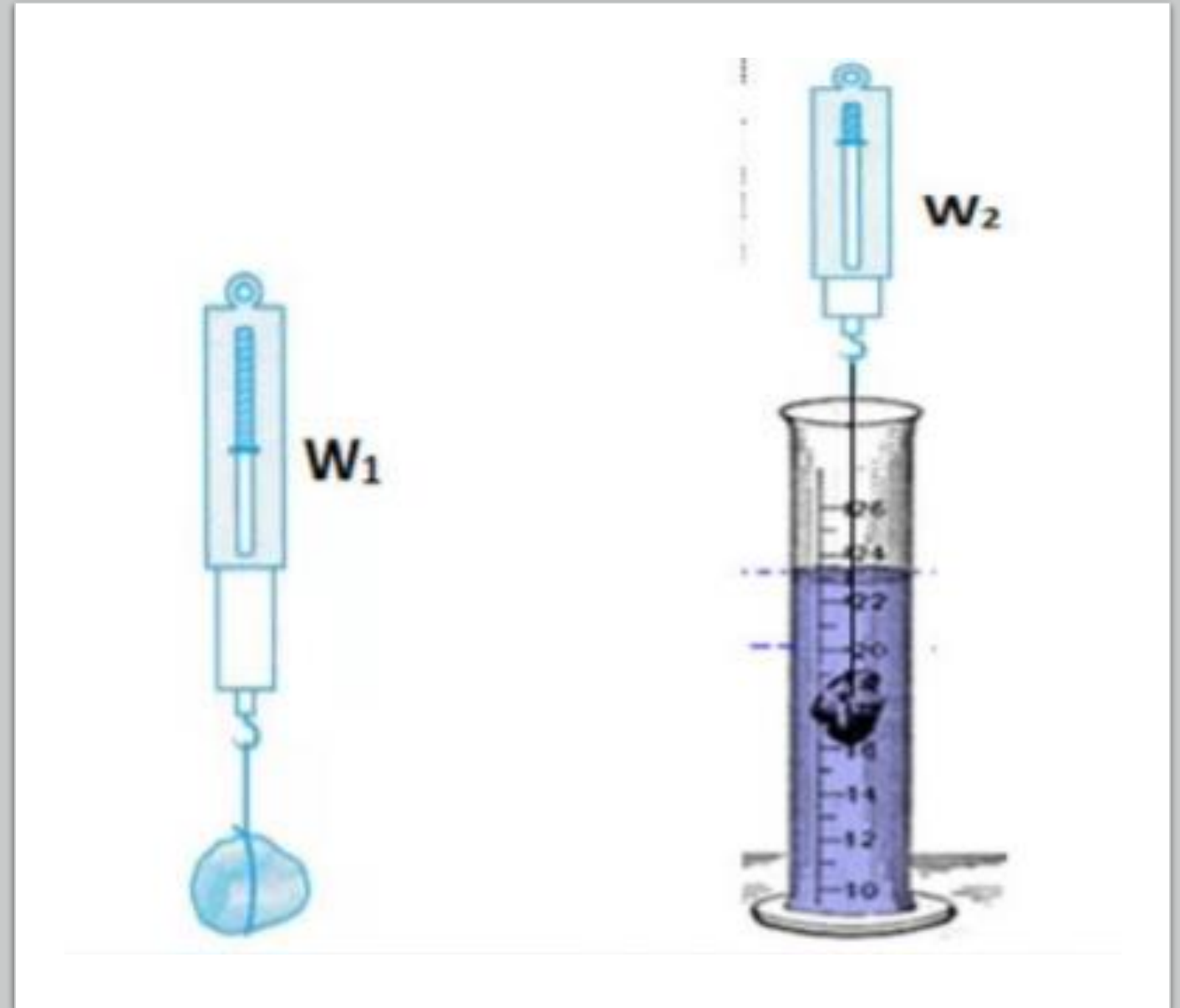
$$B_2 = W_1 - W_2$$

Where

B_2 is the buoyant force

W_1 is the weight of the block in air

W_2 is the weight of the block in fluid



Percent Difference

- To get the percent difference between the buoyant forces from the two methods, use the equation

$$\% \text{ difference} = \frac{|B_2 - B_1|}{\frac{1}{2}(B_2 + B_1)} \times 100$$

where

B_1 is the buoyant force using the fluid displacement method

B_2 is the buoyant force using the loss of mass method



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DATA AND RESULT

The following tables are to be used in gathering data from the experiment. The data will be used in the Summative Assessment.

Table 1.1 Fluid Displacement Method

QUANTITY	MEDIUM		
	WATER	KEROSENE	BRINE SOLUTION
Density of fluid (kg/m^3)	1000	820	1030
Volume of fluid displaced, $V_{\text{displaced}}$ (m^3)			
Buoyant force, B_1 , (N)			

Table 1.2 Loss of Mass Method

QUANTITY	MEDIUM		
	WATER	KEROSENE	BRINE SOLUTION
Weight of steel block in air, W_1 , (N)			
Weight of steel block in medium, W_2 , (N)			
Buoyant force, B_2 , (N)			

Table 1.3 Percent Difference of Buoyant Force

	WATER	KEROSENE	BRINE SOLUTION
% Difference			



REFERENCES

Electricalacademia, *RC Circuit Time Constant/ Charging and Discharging of Capacitor*. Retrieved from <https://electricalacademia.com/basic-electrical/rc-circuit-time-constant-rc-time-constant/>